Test #2

1) The tables of ordered pairs represent some points on the graph of line q *and* v
Which system of equations is represented by lines q *and* v?

A. $21x - y = 9$
$5x + 6y = 40$
B. $3x - y = -27$
$x + 2y = 16$
C. $21x - y = 9$
$5x + 6y = 20$
D. $9x - y = -27$
$x + 2y = 8$

Line q

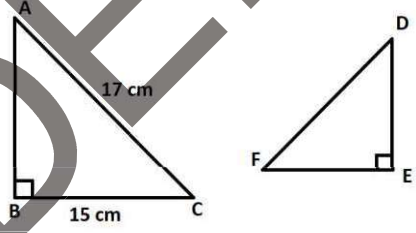
X	−9	−3	2
Y	0	18	33

Line v

X	−4	0	10
Y	10	8	3

2) Kayle was cutting right triangles from wood to one of an art project. Two of the right triangles she cut are shown below. If $\Delta ABC \sim \Delta DEF$ with right angle B and E, BC=15 cm and AC= 17 cm. what is the measure of $\angle F$ to the nearest degree?

A. 28°
B. 41°
C. 62 °
D. 88°



3) Which equation represents the line passing through the points (3, 2) and (−9, 6)?

A. $x - 3y = 9$
B. $x + 3y = 9$
C. $3x - y = -9$
D. $3x + y = 9$

3D ACADEMY

Test #2

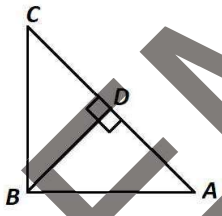
3D ACADEMY

4) Determine the number of solutions based on the graph for the following system of equations: $2x + 5y = 7$

$10y = -4x + 14$

A. Exactly one solution
B. No solutions
C. Infinitely many solutions
D. Exactly two solutions

5) In the diagram below of ΔABC, $\angle ABC$ is a right angle $AC = 12, AD = 8$ and altitude $\overline{BD}$ is drown, what is the length of $\overline{BC}$?



A. $4\sqrt{2}$
B. $4\sqrt{3}$
C. $4\sqrt{5}$
D. $4\sqrt{6}$

6) For what value(s) of x is $\frac{2x+3}{5x+3} = \frac{2}{5}$?

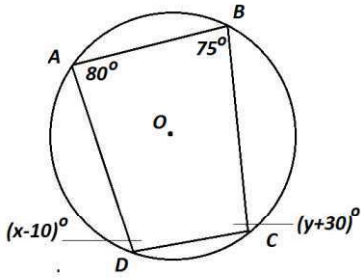
A. $x = 1$
B. $x = 3$
C. No real numbers
D. All real numbers

7) A circle whose center is the origin passes through the point $(-5,12)$ which point also lies on the circle?

A. $(10\,, 3)$
B. $(-12\,, 13)$
C. $(11\,, 2\sqrt{12}\,)$
D. $(-8\,, 5\sqrt{21})$

8) Quadrlateral ABCD is inscribed in circle O, as shown below. If $m\angle A = 80°$, $m\angle B = 75°$, $m\angle C = (y + 30)°$, and $m\angle D = (x + 10)°$ which statemet is true?

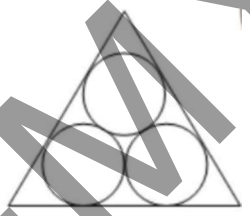
A. $x = 85\ and\ y = 50$
B. $x = 90\ and\ y = 45$
C. $x = 110\ and\ y = 75$
D. $x = 115\ and\ y = 70$



3D ACADEMY

9) Which line divides any tringle into two triangles of equal area?
   A. A median
   B. an altitude
   C. an angle bisector
   D. A perpendicular bisector of a side

10) In the figure each of the three circles is tangent to the other 2 each side of the triangle is tangent to two of the circles. Given that each circle has a radius of 4, what is the perimeter of the triangle?



   A. $8 + 8\sqrt{3}$
   B. $8 + 8\sqrt{2}$
   C. $24 + 24\sqrt{3}$
   D. $24 + 24\sqrt{2}$

11) A manufacturing company is building a rectangular room in their warehouse to store their products. The length of the room is 1 more than 3 times its width. The area of the rooms is 80 square meters. What are the dimensions of the room?
   A. $w = 4\,m\,, l = 20m$
   B. $w = 5m\,, l = 16m$
   C. $w = 5\frac{1}{3}m\,, l = 15m$
   D. $w = 8m\,, l = 10m$

12) A bag contains 16 red checkers and 16 black checkers. Anne removes 1 black checker and 2 red checkers from the bag and does not replace them. She then reaches into the bag and selects a checker at random. What is the probability that the checker is red ?
   A. $\frac{1}{2}$
   B. $\frac{14}{15}$
   C. $\frac{14}{29}$
   D. $\frac{15}{29}$

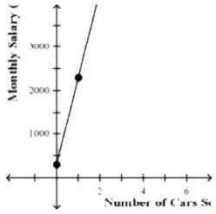
13) Jasmine and her sister are saving to buy MP3 plyers. Jasmine has $50 and plans to save $10 per week. Her sister has $80 and planes to save $7 per week. In how many weeks will Jasmine have more money saved than her sister?
   A. 2 weeks
   B. 4 weeks
   C. 10 weeks
   D. 11 weeks

3D ACADEMY

14) Thomas is a car salesman. The table shows the monthly salary that Thomas earns for the number of cars he sells. Use the data to graph the linear function. Write the equation of the line, identify the slope of the line and explain what the slope means.

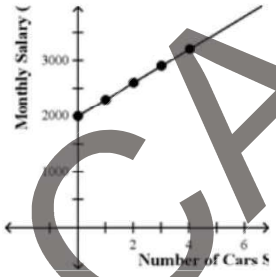
Number of cars sold	Monthly salary Earned
0	$2000
1	$2300
2	$2600
3	$2900
4	$3200

(A)



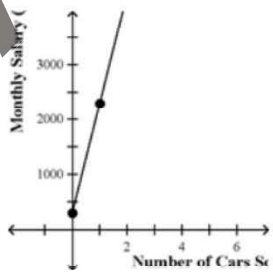
$y = 2000x + 300$; The slope of the line is 300.
This means that for every car Thomas sells, he earns and additional 300 $

(B)



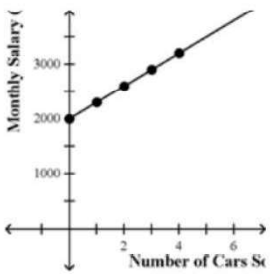
$y = 300x + 2000$; The slope of the line is 300.
This means that for every car Thomas sells, he earns an additional 300 $

(C)



$y = 2000x + 300$; The slope of the line is 2000.
This means that for every car Thomas sells, he earns $ 2000.

(D)



3D ACADEMY

Test #2

3D ACADEMY

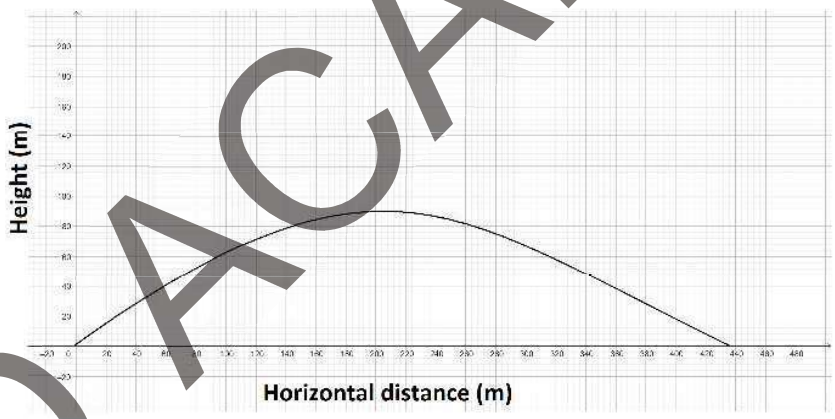
15) If x is a positive odd integer and y is a negative odd integer such that $|x| \neq |y|$ which of the following must produce a positive even integer?

A. y^x
B. x^y
C. $x - y$
D. $\frac{y^2}{x}$

16) One side of an equilateral triangle measures 6 inches. What is the height of the triangle?

A. $\sqrt{3}$
B. $3\sqrt{3}$
C. 6
D. $6\sqrt{3}$

17) The trajectory of a potato launched from a potato cannon on the ground at an angle of 45 degrees with an initial speed of 65 meters per second can be modeled by the parabola $f(x) = x - 0.0023x^2$ where the x-axis is the ground. Find the height of the highest point of the trajectory and the horizontal distance the potato travels before hitting the ground.

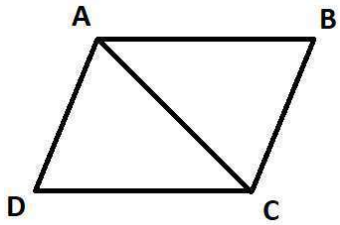


A. height: 109 m, distance: 435 m
B. height: 121 m, distance: 418 m
C. height: 118 m; distance: 421 m
D. height: 102 m; distance: 409 m

3D ACADEMY

Test #2

18) ABCD is a rhombus. How do you complete the explanation that states why $\Delta ABC \cong \Delta CDA$?

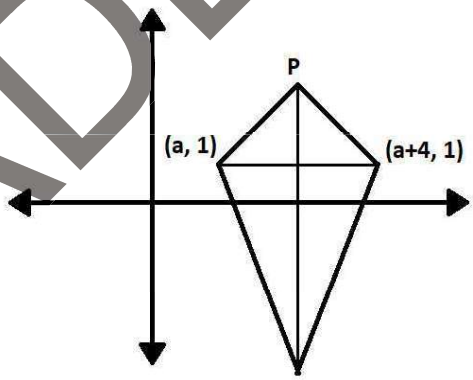


$\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$ by the definition of rhombus. $\overline{AC} \cong \overline{AC}$ by the Reflexive Property of Congruence, so $\Delta ABC \cong \Delta CDA$ by…….

A. ASA
B. AAS
C. SAS
D. SSS

19) The figure shown is kite. What is the x-coordinate of point P ?

A. $\frac{a}{2} + 4$
B. $\frac{a+4}{2}$
C. $a + 2$
D. $2a + 4$



20) The solution of this system of inequalities forms a triangular region

$y \leq 3x$

$y \leq -\frac{1}{2}x + 4$

$y \geq 0$

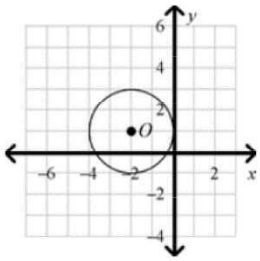
What is the area, in square units, of the region?

A. $2\frac{2}{7}$
B. $4\frac{4}{7}$
C. $13\frac{5}{7}$
D. $27\frac{3}{7}$

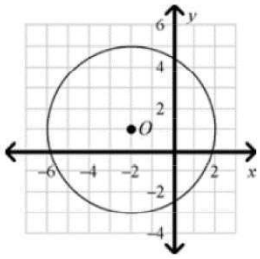
3D ACADEMY

21) A manufacturing is designing a two-wheeled cart that can maneuver through tight spaces. On one test model, the wheel placement (Center) and radius is modeled by the equation $(x+2)^2+(y-1)^2=4$ what is the graph that shows the position and radius of the wheels?

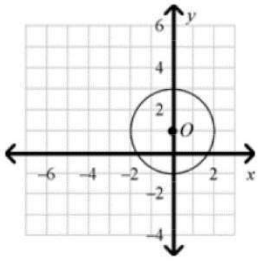
(A)



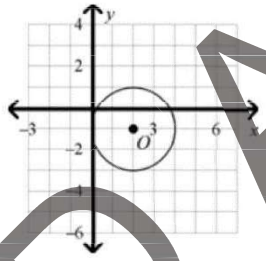
(B)



(C)



(D)



22) What is the length of the diagonal of a rectangle whose perimeter is 34 and whose area is 60 ?

A. 69
B. 49
C. 26
D. 13

3D ACADEMY

23) A system of equations is shown below

Equation A: $5x + 9y = 12$

Equation B: $4x - 3y = 8$

Which method eliminates one of the variables?

A. Multiply equation A by $\frac{1}{3}$ and subtract the result from equation B.

B. Multiply equation B by 3 and add the result to equation A.

C. Multiply equation A by 2 and equation B by -3 and add the results together.

D. Multiply equation B by 5 and equation A by 4 and add the results together.

24) If g is a function defined over the set of all real numbers and $g(x - 1) = 3x^2 + 5x - 7$ then which of the following defines $g(x)$?

A. $g(x) = 3x^2 - x - 9$

B. $g(x) = 3x^2 + 5x + 1$

C. $g(x) = 3x^2 + 11x + 1$

D. $g(x) = 3x^2 + 11x - 6$

25) Which of the following complex numbers is a sum of $\sqrt{-48}$ and $\sqrt{-27}$?

A. $-5\sqrt{3}$

B. $-7\sqrt{3}$

C. $5i\sqrt{3}$

D. $7i\sqrt{3}$

26) Given the values of x , $f(x)$ *and* $g(x)$ in the table below, what is the value of $g(f(2))$?

A. 2

B. 3

C. 6

D. 9

x	$f(x)$	$g(x)$
0	0	-5
1	3	-2
2	6	1
3	9	2
4	12	3
5	15	7
6	18	9

3D ACADEMY

Test #2

3D ACADEMY

27) Rita, a junior, is Running for president of the key club. There are 9 other juniors running for the same position. If, historically, a junior has a 1 in 3 chance of being elected presides of the club, what are Rita's chances of becoming president of the key club?

A. $\frac{1}{3}$
B. $\frac{1}{9}$
C. $\frac{1}{10}$
D. $\frac{1}{30}$

28) What is the value of $f(x) = \frac{x^3}{4}$, when $x = \frac{1}{2}$

A. $\frac{1}{32}$
B. $\frac{1}{16}$
C. $\frac{1}{12}$
D. $\frac{1}{8}$

29) For winning a chess tournament, the captain Ayleen, will receive one-third of what is left of a $500 prize after the team's coach gets $20. What remains after Ayleen gets her share will be split evenly among the other team members. If the team has a total of 5 players, how much will Ayleen's teammate Brenda receive?

A. $40
B. $80
C. $96
D. $100

30) The stem and leaf plot below depicts the daily average temperature (rounded to the nearest degree) for Trees Ville over a two-week period with weekend leafs - W. X. Y, and Z . – absent. If the median daily temperature was 65°, what could be the mode of the data set ?

A. 59°
B. 64°
C. 66°
D. 77°

Stem	leaf
5	8, w, 9
6	X, 2, 4, 4, Y, 6, 6, 9
7	7, 7
8	Z

3D ACADEMY

Test #2

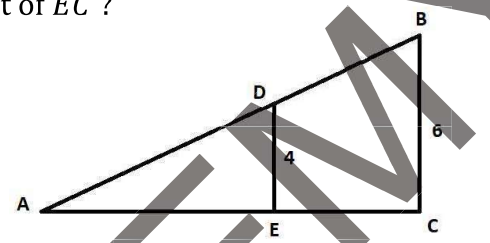
3D ACADEMY

31) A circle has a circumference of 36π, what is the area?

A. $12\,\pi$
B. $18\,\pi$
C. $81\,\pi$
D. $324\,\pi$

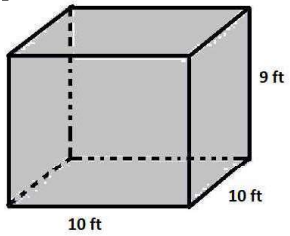
32) In the diagram shown $DE \parallel BC$, DE measures 4 units, BC measures 6 units, and AC measures 9 units. What is the measurement of EC ?

A. 3
B. 5
C. 6
D. $\frac{27}{2}$



33) Rebecca is loading medical supply bones into a crate. Each supply box is 1.5 feet tall, 1 foot wide, and 2 feet deep. The crate is 9 feet high, 10 feet wide, and 10 feet deep. What is the maximum number of supply boxes can she pack in the crate?

A. 200
B. 300
C. 450
D. 600



34) Let $f(x) = \begin{cases} \sqrt{2x-3} - x, x > 5 \\ 1 - 2x \quad\quad , x \leq 5 \end{cases}$ compute $f(6)$?

A. -11
B. -6
C. -3
D. 3

3D ACADEMY

Test #2

35) $f(x) = x^2 - 3x$ what is the value of $f(5)$?

A. 4
B. 8
C. 10
D. 16

36) Triangle JGR is similer to triangle MST which statement is not always true ?

A. $\angle J \cong \angle M$
B. $\angle G \cong \angle T$
C. $\angle R \cong \angle T$
D. $\angle G \cong \angle S$

37) Pedro buys 60 baseball cards to start his collection. Each week, he adds 5 cards to his collection. Lea begins her collection with 80 cards and adds 3 cards to her collection each week. Which of the following equations, when solved, gives the number of weeks, t, after starting their collections that pedro and lea will have the same number of cards in their collection?

A. $60 + 3t = 80 + 5t$
B. $60 + 5t = 80 + 3t$
C. $60 - 3t = 80 + 5t$
D. $60 - 5t = 80 + 3t$

38) Jade has test scores of 82, 75, amd 78. What does she need to score on her next test so that she has an average test score of 80?

A. 75
B. 78
C. 80
D. 85

39) What happens to the graph of $y = |x|$ when the equation changes to $y = |x - 2|$?

A. The graph shefts up 2 units.
B. The graph shefts left 2 units.
C. The graph shefts right 2 units.
D. The graph shefts down 2 units.

3D ACADEMY

Test #2

40) The frequency table shows the prizes of tickets in a school raffle. There are 1,000 total ticket for the raffle. A local business purchased 100 tickets, How many free movie tickets should the business expect to win ?

A. 10 Movie tickets
B. 20 Movie tickets
C. 50 Movie tickets
D. 100 Movie tickets

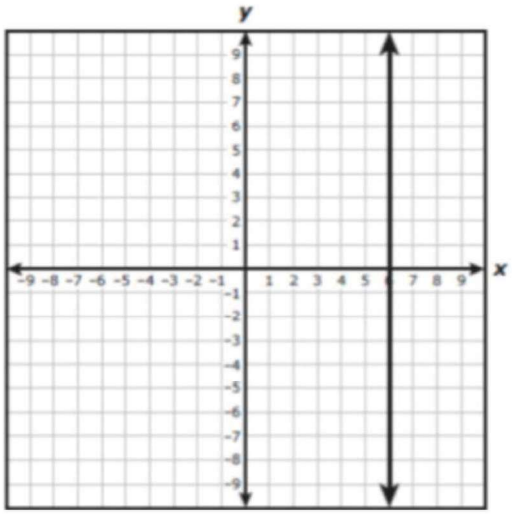
Prize	Number of winning tickets
$500	1
$30	9
$10	90
$5	400
Free movie ticket	500

41) A furnace technician charges $20 per hour and a one-time service charge of $30. Which equation represents the total charge (y) for working (x) hours?

A. $y = 20x + 30$
B. $y = 30x + 20$
C. $y = 20(x - 30)$
D. $y = 20(x + 30)$

42) What is the equation and slope of the line shown in the grid?

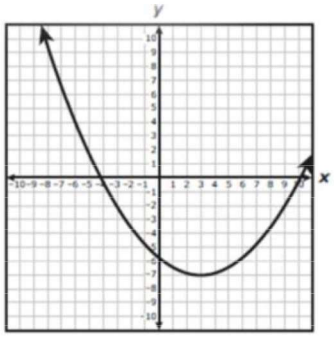
A. $y = 6$; $slope\ is -\frac{1}{6}$
B. $x = 6$; $slope\ is\ zero$
C. $y = 6$; $slope\ is\ 6$
D. $x = 6$; $slope\ is\ undifined$



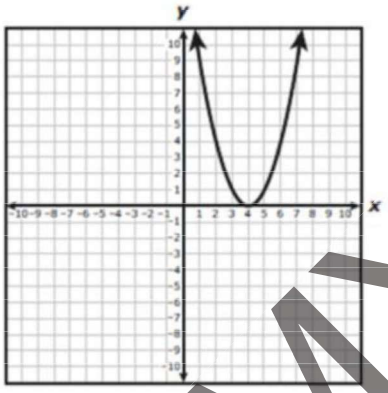
3D ACADEMY

43) Which graph best represents a quadratic function that has only one zero?

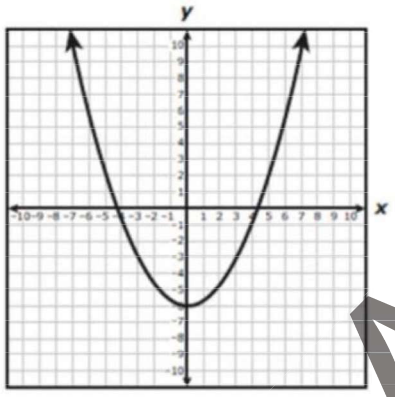
A



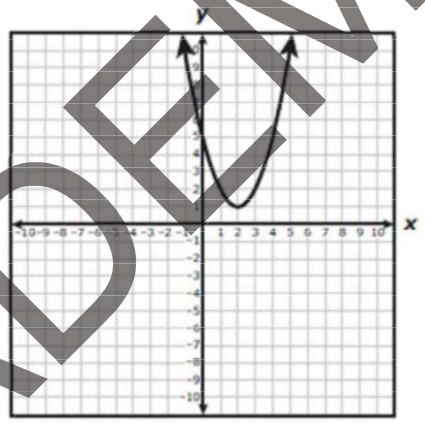
B



C



D



44) There are 15 plates in a kitchen cabinet. The diameter of each is either 7 inches or 12 inches. The diameter of all 15 plates compined is 14 inches. Which system of equations can be used to find the number of 7-inch plates, x, and the number of 12-inch plates, y, that are in the cabinet?

A. $x + y = 140$
$12x + 7y = 15$

B. $7x + 12y = 140$
$7x + 12y = 15$

C. $x + y = 15$
$7x + 12y = 140$

D. $x + y = 15$
$12x + 7y = 140$

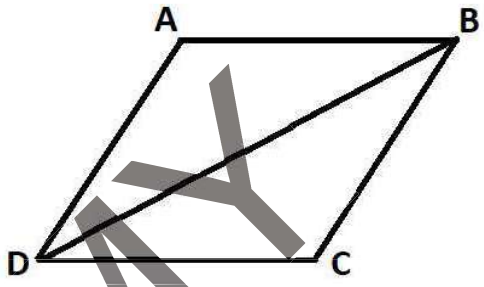
3D ACADEMY

Test #2

3D ACADEMY

45) What is the missing reason for the proof?
Given: parallelogram $ABCD$ with diagonal $\overline{BD}$ proof $\Delta ABD \cong \Delta CDB$

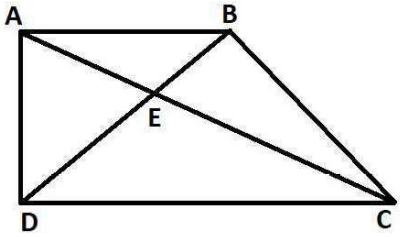
Statement	Reasons
1. $\overline{AD} \cong \overline{BC}$	1. Definition of a parallelogram
2. $\angle ADB \cong \angle CBD$	2. alternative interior angles theorem
3. $\overline{AB} \parallel \overline{CD}$	3. Definition of a parallelogram
4. $\angle ABD \cong \angle CDB$	4. alternative interior angles theorem
5. $\overline{DB} \parallel \overline{DB}$	5. reflexive property of conguence
6. $\Delta ABD \cong \Delta CDB$	6. ??



A. reflexive property of conguence
B. ASA
C. alternative interior angles theorem
D. SSS

46) In trapezoid ABCD below, $\overline{AB} \parallel \overline{CD}$
If $\overline{AE} = 5.2$, $\overline{AC} = 11.7$ *and* $CD = 10.5$, what is the length of $\overline{AB}$ to the nearest tenth?

A. 4.7
B. 6.5
C. 8.4
D. 13.1



47) In parallelogram $ABCD$, diagonals $\overline{AC}$ *and* $\overline{BD}$ intersect at E. Which statement proves $ABCD$ is a rectangle ?

A. $\overline{AC} \cong \overline{BD}$
B. $\overline{AB} \perp \overline{BD}$
C. $\overline{AC} \perp \overline{BD}$
D. $\overline{AC}$ bisects $\angle BCD$

3D ACADEMY

48) Which of the following phrases describes the graph of the line $y = 5$?

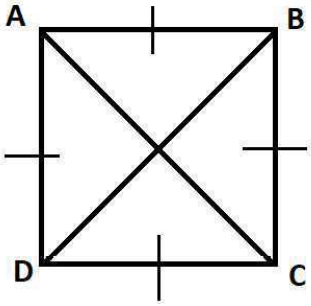
A. Horizontal line through quadrant I and quadrant II.
B. Vertical line through quadrant I and quadrant II.
C. Horizontal line through quadrant I and quadrant IV.
D. Vertical line through quadrant I and quadrant IV.

49) If the radii of two circles are equal, then the circles are congruent what is the inverse of the statement above ?

A. If the radii of two circles are equal, then the circles are not congruent.
B. if two circles are not congruent, then the radii are unequal.
C. If two circles are congruent, then the radii of the circles are equal.
D. If the radii of two circles are unequal, then the circles are not congruent.

50) Given ABCD as marked ,what additional information whould help you to prove that ABCD is a square ?

A. $\overline{AC} \cong \overline{BD}$
B. $\overline{AC} \perp \overline{BD}$
C. $\overline{AC}\ and\ \overline{BD}$ bisect each other.
D. $\overline{AC}$ bisect $\angle DAB\ and\ \angle BCD$, $\overline{BD}$ bisect $\angle ABC\ and\ \angle CDA$



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Test #2

3D ACADEMY

Answer Keys:

Question	Answer	Question	Answer
1	B	26	D
2	A	27	D
3	B	28	A
4	C	29	B
5	B	30	C
6	C	31	D
7	C	32	A
8	D	33	B
9	A	34	C
10	C	35	C
11	B	36	B
12	C	37	B
13	D	38	D
14	B	39	C
15	C	40	C
16	B	41	A
17	A	42	D
18	D	43	B
19	C	44	C
20	C	45	B
21	A	46	C
22	D	47	A
23	B	48	A
24	C	49	D
25	D	50	B

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